

#Class-01 (09.02.2026)

Corpus → large collection of computer readable text

↑ plural

Corpora

Normalization

Age

min-max scalar

(0-1)

5 →

10 →

15 →

Annotation

NLP Components

Text Normalization

Sentence Segmentation: Given a text, we need to separate each sentences

Challenges:

Solution:

Rules: → very much difficult manually

Machine learning: → easier

Tokenization

Challenges:

Solution: →

morphemes → smallest unit of language that carries its own meaning.

word piece tokenization } advanced
Byte pair Encoding } algo used
by GPT + others

vocabulary reduce

root word

↓
"dog" "dogs"

connect
} ing
} tion
} ed

stemming → rule based

porter stemmer → 1980 (string matching)

↳ fast but inaccurate

↓
equivalence → ~~equival~~

Lemmatization:

hand built lexicon
list of words

→
{ 'running': 'run',
 'runner': 'run',
 'ran': 'run',
 ...
}

Accurate but slower + chicken-egg
scenario with parts
of speech tagging

Embedding: technique → process a word or phrase to a numerical vectors.

NLP Annotation → Associating extra information to a piece of text.

→ **Pos Tag:** → Assigning grammatical categories for words

→ Penn TreeBank Tags

→ Universal POS Tags

Tags →

phrase $\rightarrow$ verb 2778 27

She liked it very much.
pronoun verb pron. adv punctuation

Challenges:

ambiguity

I Painted the room

The Painted room

Sequence Track $\leftarrow$ Soln

$\rightarrow$ Named Entity Recognition (NER)

$\hookrightarrow$ Identify phrases that are named people, location, org. etc $\hookrightarrow$ Tags

Common NER Tags:

person $\rightarrow$ PER, location $\rightarrow$ LOC, organization $\rightarrow$ ORG

Geo Political Entity $\rightarrow$ GPE, Creative/Artwork $\rightarrow$ ART

① BIO
 $\hookrightarrow$ outside
 $\hookrightarrow$ Inside
 $\hookrightarrow$ Begin

② BILOU
 $\hookrightarrow$ Unit (single word entity)
 $\hookrightarrow$ outside
 $\hookrightarrow$ Last

eg; medical history

Hugging Face

challenge → ambiguity

- Washington was born into slavery
<PER>
- Washington went up 2 games to 1.
<ORG>
- Blair arrived in Washington today.
<LOC>
- Washington passed a primary seatbelt law.
<GPE>

Dr Jenifer Smith

<B-per> <I-per> <

BIO;

Mr Adib Hossain and Kazi Hasan is going out for

P P P O P P O O O O
B-P I-P I-P O B-P I-P O O O O

a meeting in Badda.

O O O L O
O O O B-L O